

# Alyssa Dang

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## EDUCATION

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**Carnegie Mellon University, Pittsburgh, PA**

*Expected May 2028*

*B.S. in Human-Computer Interaction + Cognitive Science (Machine Learning) | GPA: 4.0/4.0*

- Relevant Coursework: Computational Neuroscience, Designing Human-Centered Software (in progress), Cognitive Psychology, Fundamentals of Programming

## RESEARCH & TECHNICAL EXPERIENCE

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**Incoming Research Assistant, Laboratory of Computational Perception**

*Starting Aug 2026*

- Training deep neural networks in PyTorch on a range of auditory tasks to study how task objectives shape the models' learned internal representations

**V1 Neural Decoding & Systems Identification, Computational Neuroscience Lab**

*Apr 2026*

- Analyzed mouse primary visual cortex (V1) Neuropixel recordings (Allen Institute/DANDI) across drifting-grating and Gabor stimuli using Gaussian-kernel smoothing, tuning curves, spatial receptive field mapping, and Linear-Nonlinear-Poisson (LNP) model validation via spike-triggered averaging
- Applied Principal Component Analysis (PCA) via SVD on zero-centered population response matrices, identifying a low-dimensional manifold where 6 principal components captured 95% of variance
- Built OLS linear regression decoders with circular target encodings (80/20 train/test split), demonstrating significantly higher decoding accuracy for stimulus orientation over direction (31° vs. 67° mean test error)

**LTP/LTD Circuit Simulation, Computational Neuroscience Lab**

*Feb 2026*

- Built a Leaky Integrate-and-Fire (LIF) circuit model in Python (NumPy, Matplotlib, Jupyter Notebook) simulating AMPA/NMDA receptor kinetics, calcium current, and membrane voltage dynamics of an excitatory synapse
- Implemented Spike-Timing-Dependent Plasticity (STDP) via calcium-threshold learning rules modulating synaptic conductance (LTP/LTD) within a biologically accurate 20ms spike-timing window
- Validated model behavior against literature by plotting and analyzing I-V curves and conductance history

**NBA Outcome Prediction & Optimization Model, Machine Learning & Probabilistic Modeling Project**

*Apr 2026*

- Built Random Forest classification models in R to predict NBA home win probabilities using team performance data across multiple model architectures (all predictor variables, top predictors, and home team predictors)
- Evaluated models via 5-fold cross-validation, achieving a 0.102 Brier score on top feature sets
- Quantified prediction uncertainty via tree standard deviations and mapped win probabilities against market break-even odds to guide expected-value decision-making

**Product & UI/UX Designer, CravingChef (iOS App Store)**

*Jul 2026 – Present*

- Synthesized user feedback and behavioral analytics into interface decisions, establishing human-centered research workflows applicable to neurotechnology and assistive human-machine interfaces
- Designed and usability-tested low-to-high fidelity UI flows and prototypes across product surfaces in Figma

## SKILLS

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**Technical:** Python, PyTorch, NumPy, Matplotlib, R, Dimensionality Reduction (PCA, SVD), Linear Regression Decoders, LNP Modeling, LIF Neurons, Spike-Triggered Averaging, Jupyter, HTML/CSS, JavaScript, Godot, AI-Driven Workflow

**Research & Signal Processing:** Neural Data Processing (Neuropixels/DANDI), Signal Filtering, Receptive Field Mapping, Computational Modeling, Cross-Validation, Usability Testing

**Design Tools:** Figma, Adobe Creative Suite, Google Workspace